



Document Management System



Sultanate of Oman
Ministry of Health
The Royal Hospital
Department of Child Health

Code: CH-MG-PRT-3-Vers.1.0

Effective Date: 12/10/2023

Target Review Date: 10/10/2026

Title: Galsulfase (Naglazyme) Protocol

Introduction:

Naglazyme (Purified galsulfase), a recombinant form of human N-acetylgalactosamine 4-sulfatase, is a glycoprotein with a molecular weight of approximately 56 kD. The rationale for enzyme replacement therapy is to restore a level of enzymatic activity sufficient to hydrolyze the accumulated substrate and to prevent further accumulation. After intravenous infusion, galsulfase is rapidly removed from the circulation and taken up by cells into lysosomes, most likely via mannose-6 phosphate receptors. It is FDA approved medication in 2005 and it has been shown to be safe and effective in the treatment of MPS VI.

Indication:

Patients with a confirmed diagnosis of Mucopolysaccharidosis VI (MPS VI; N-acetylgalactosamine 4-sulfatase deficiency; Maroteaux-Lamy syndrome) by enzyme assay or molecular testing

About MPS VI:

Mucopolysaccharidosis type VI (MPS VI, Maroteaux-Lamy syndrome) is an autosomal recessive lysosomal storage disorder, characterized primarily by skeletal dysplasia and joint contracture and cardio-pulmonary involvement. Patients have reduced exercise capacity and endurance, and limitations in joint range of motion. Patients with MPS VI has a deficiency of specific lysosomal enzymes required for the catabolism of glycosaminoglycans (GAGs). The accumulation of GAGs causes progressive cellular, multi-system damage, organ failure, and reduced life expectancy.

Treatment protocol:

1-Dose regimen: at 1 or 2 mg/kg/week

2-Pre-treatment: Pre-treatment with antihistamines, with or without antipyretics, is recommended 30—60 minutes prior to the start of the infusion to minimize the potential occurrence of infusion-associated reactions (IARs)

3-Method of administration: The initial infusion rate is adjusted so that approximately 2.5% of the total solution is infused during the first hour, with infusion of the remaining volume (approximately 97.5%) over the next 3 hours.

-100 ml infusion bags should be considered for patients who are susceptible to fluid volume overload and weigh less than 20 kg; in this case the infusion rate (ml/min) should be decreased so that the total duration remains no less than 4 hours.

- Severe or life-threatening hypersensitivity to the active substance or to any of the excipients, if hypersensitivity is not controllable.
- Caution must be exercised in the management and treatment of patients with compromised airways by limitation or careful monitoring of antihistamine and other sedative medicinal product use.
- Patients who present with an acute febrile or respiratory illness may need to have their Naglazyme infusions delayed.

5- Management of infusion-associated reactions

Based on data obtained during Naglazyme clinical trials, the majority of patients are expected to develop IgG antibodies to galsulfase within 4-8 weeks of treatment initiation.

In the Naglazyme clinical trials, IARs were usually manageable by interrupting or slowing the rate of

Infusion and by (pre-) treating the patient with antihistamines and/or antipyretics (paracetamol), thus enabling the patient to continue treatment.

As there is little experience on resumption of treatment following prolonged interruption, caution is to be used due to the theoretical increased risk of hypersensitivity reaction.

Monitoring & Follow up:

Naglazyme-treated patients experienced 90% improvement in

1. the distance walked in 12 minutes
2. experienced a 5.7 stair per minute improvement in the 3 Minute Stair Climb
3. a mean decrease in urinary glycosaminoglycan excretion
4. Improvement in pulmonary function

- Regular follow with cardiology for MV, orthopedics and ophthalmologist annually

References:

-Recommendations for the management of MPS VI: systematic evidence- and

Consensus-based guidance, Mehmet Umut Akyol¹, Tord D. Alden², Hernan Amartino³ et al, 2019

- NAGLAZYME® (galsulfase) DOSING & ADMINISTRATION GUIDE,

(Galsulfase), Summary of Product Characteristics.

- **Management Guidelines for Mucopolysaccharidosis VI, Roberto Giugliani, Paul Harmatz and James E. Wraith, Pediatrics August 2007**

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